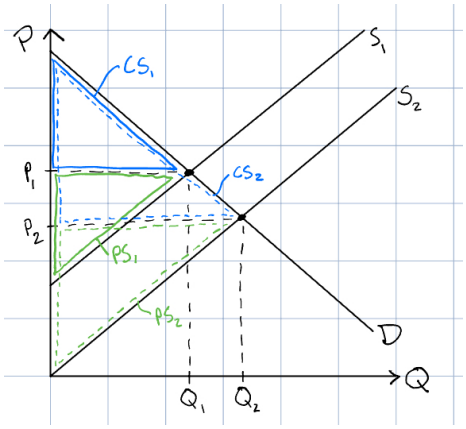
HW 7.4 Solns

Principles of Microeconomics

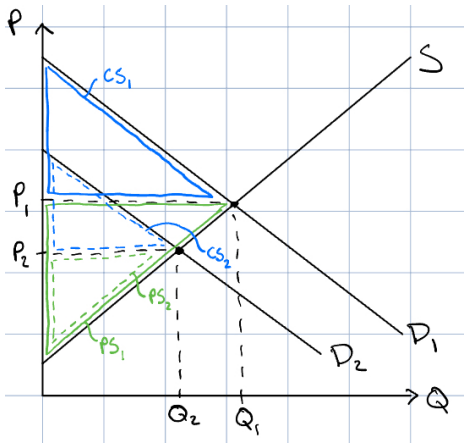
October 15, 2025

Ch. 7 #9, 10

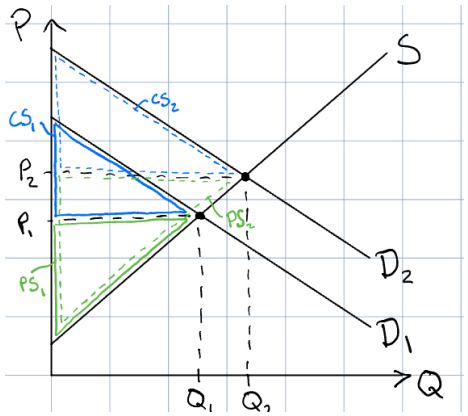
9. (a) The advancement in production technology will shift the supply curve out. That increases the equilibrium quantity and decreases the equilibrium price. As a result, consumer surplus and producer surplus both increase.



(b) That makes typewriters and computers substitutes. The decreased price of computers shifted demand for typewriters left. That decreases the equilibrium quantity and price. As a result, consumer surplus and producer surplus both decrease. Typewriter producers should be sad about the technological advance because it decreases the value they accrue in the typewriter market.



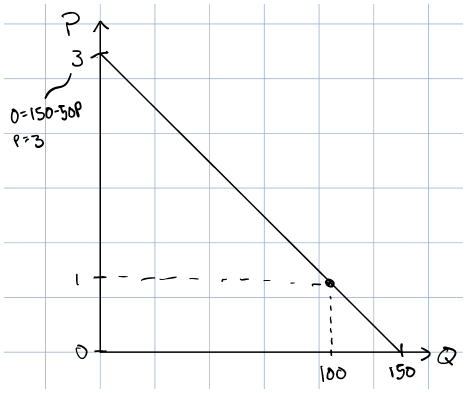
(c) Computers and software are complements, so the decreased price of computers increases demand for software. That increases the equilibrium quantity and price of software. As a result, consumer surplus and producer surplus both increase. Software producers should be happy about the technological advance because it increases the value they accrue in the software market.



(d) It does. We see that software producers accrued a lot more surplus when computers became more widespread.

10. (a) Provider A: \$0 because each additional movie comes at no additional charge.
Provider B: \$1 because each additional movie comes at a charge of \$1.

(b) The friend should buy until their willingness to pay for an additional movie equals the cost of an additional movie.



Provider A: $Q_D = 150 - 50(0) = 150$
Provider B: $Q_D = 150 - 50(1) = 100$

(c) She would pay Provider A the one time flat fee of \$120. She would pay Provider B \$1 per move, \$100.

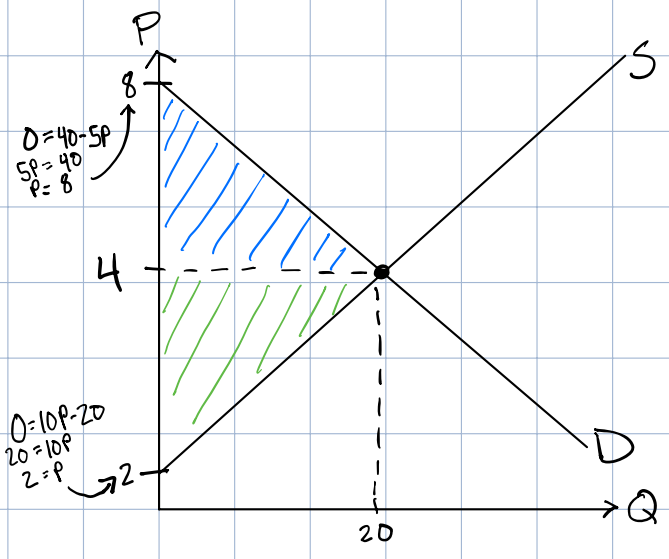
(d) Provider A: (We have to think about price a little differently here)
Value accrued from movies = $\frac{1}{2}(3)(150) = 225$
Total spent on movies = \$120
Consumer Surplus = $225 - 120 =$ \$105
Provider B:
Consumer Surplus = $\frac{1}{2}(3-1)(100) =$ \$100

(e) I would recommend Provider A because they would get more consumer surplus from streaming through Provider A.

Non-Mankiw #1, 2

Solns

1. a.



$40-5P^* = 10P^* - 20$

$60 = 15P^*$

$4 = P^*$

$Q^* = 40-5(4)$

$= 20$

b. $CS = \frac{1}{2}(8-4)(20) = 40$

$PS = \frac{1}{2}(4-2)(20) = 20$

2. a. $105-15P^* = 25P^* - 15$

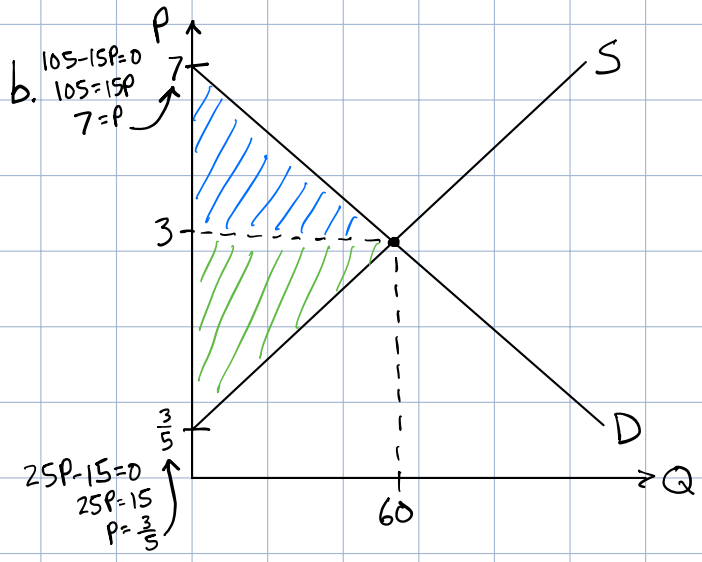
$120 = 40P^*$

$3 = P^*$

$Q^* = 105-15(3)$

$= 60$

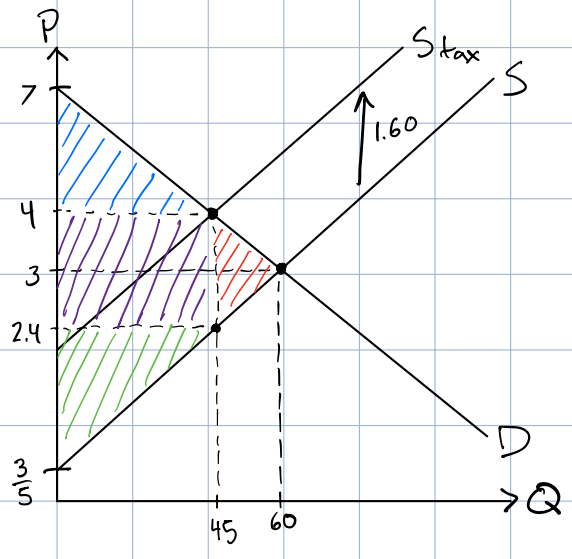
b.



$CS = \frac{1}{2}(7-3)(60) = 120$

$PS = \frac{1}{2}(3-\frac{3}{5})(60) = \frac{1}{2}(\frac{12}{5})(60) = 72$

c.



$105-15P_{consumer} = 25(P_{consumer}-1.60)-15$

$105-15P_c = 25P_c - 40 - 15$

$160 = 40P_c$

$4 = P_c$

$P_{supplier} = 4-1.6$

$= 2.4$

$Q_{tax} = 105-15(4)$

$= 45$

d. $P_{consumer} = 4$ $P_{producer} = 2.4$ (See work above)

e. $CS = \frac{1}{2}(7-4)(45) = \frac{135}{2}$

$PS = \frac{1}{2}(2.4-0.6)(45) = \frac{1}{2}\left(\frac{12}{5}-\frac{3}{5}\right)(45) = \frac{9}{5}\cdot\frac{45}{2} = \frac{81}{2}$

f. $Tax\ Rev = 1.6(45) = 72$

g. $DWL = \frac{1}{2}(4-2.4)(15) = 0.8(15) = 12$